

Art

Artists have traditionally sought to express themselves using a variety of media. Classes of artistic expression such as the symphony orchestra, visual arts, dance, and motion pictures each have their own subforms of materials used to forge the artworks. Within the visual arts, for example, charcoal, oil and canvas, clay, steel, and even ice have been utilized as conduits for ideas. Each of these mediums has its own possibilities, quirks, and limitations.

Artists approach different media for different reasons and from different perspectives. Some artists seek to find the medium best suited to their message, while others seek to explore and extend media in different and untried ways. As a medium, virtual reality has been ripe with the possibilities for exploration and discovery.

Through exploration, artists have helped discover new uses for technology, and lead others to see possibilities beyond those originally intended and imagined. Artists are generally free of the preconceptions that constrain the creative application of new technology and are able to let their imagination soar, to show the world a glimpse of what *could* be, rather than focusing on the limitations of the medium. As a result, even for someone not interested in the application of VR as art, there is still much to learn by examining the work in this area.

In general, *art* (as it concerns exploration by artists) helps to drive public perception of a new medium. The pursuit of artistic expression uncovers interesting and sometimes broadly useful ways to interface with and make use of the medium.

This chapter discusses several uses of VR designed as artistic expression. By exploring these applications, many potential applications can be imagined. Virtual reality practitioners are only beginning to realize the potential of the medium for artistic expression, and the applications discussed in this section show a very small glimpse of the power of this medium.

8.1 AREAS OF APPLICATION

As with all the chapters on applications, putting each application into a specific subcategory does not always seem appropriate. This is particularly apparent in a chapter on art, which generally defies categorization. However, it is convenient to divide applications into similar groups for easier reference. This division of groups is not necessarily a mutually exclusive categorization.

Five categories of art applications include: *design and sketching*, *exploration of the medium*, *empathetic experiences*, and *exploration of the human condition*. A fifth category, *performance VR*, is not represented by a review of any particular applications in this book.

8.1.1 Design and sketching

The use of VR for *design and sketching* allows one to create works of art while immersed in a virtual reality environment. These works of art may be manifested as purely digital virtual worlds that must be experienced while immersed in VR, or they can be created in VR and then manifested in some other medium. That is, an architectural design may be made while immersed, yet ultimately realized as a physical building or as hardcopy two-dimensional still images. Likewise, sculpture can be created and enjoyed in the digital realm, or be taken to the physical realm by using technologies such as *stereolithography*, or by a *milling machine*.



FIGURE 8-1

ShadowLight, created by Kalev Leetaru at the National Center for Supercomputing Applications, allows architecture students to freely create full-scale architectural designs while immersed in the CAVE environment. *ShadowLight* offers many tools for editing, texturing, and creating polygons, including those commonly found in traditional two-dimensional drawing packages.

Note that there is no restriction to creating artworks that are visual in nature. Musical compositions and other sound art may be created in VR. When one considers creating art in VR, one might first think of an immersive, three-dimensional extension of standard computer drawing packages, such as typical desktop draw, paint, and modeling packages. To think this way limits the possibilities that are present in VR for dynamics in the world physics, dynamic point of view, multiperson interactivity and other forms of interactivity, and many more. One example of an art-creation program in VR that, at first blush, resembles desktop drawing tools is NCSA *ShadowLight*.

ShadowLight allows participants to create and place arbitrary polygons, paint and texture them, group objects, and use lighting effects, and provides many other drawing tools. However, it also exploits the interactive nature and collaborative nature inherent in VR. *ShadowLight* has been used as a sketching program for a design course in the University of Illinois Architecture Department, where Professor Joy Malnar believed it was important for students to sketch their ideas, at full scale, while immersed in the object they are creating. It was also used by middle school students to create their own virtual worlds in cooperation with a project called VR Savvy.

The Body Language User Interface (BLUI), from the Arctic Region Supercomputing Center, allows participants to create very fluid sculpture while immersed in a CAVE or other projection VR device. *BLUI* offers a very intuitive way to create digital sculpture. The resulting sculptures can be experienced in VR, or they can be printed out using three-dimensional prototyping printers to create physical sculptures that can be experienced in the manner to which people are accustomed to interacting with sculpture in the physical realm. *CavePainting* from Brown University allows people to freely brush strokes of virtual paint in a VR environment to create very fluid, three-dimensional virtual worlds. The paintbrushes are the analog of paintbrushes in two-dimensional computer paint programs.

This application is very interesting in its use of props as its sole user interface. That is, you use an actual paintbrush, and an actual paint bucket, etc., and are able to use techniques such as brushing and throwing paint while creating your artwork.

8.1.2 Exploration of the medium

Exploration of the medium might be considered “art for art’s sake.” It consists of artworks that do something different in order to appraise the end result. For an established medium, it might be those efforts that challenge the more traditional fare.

Many very early VR applications took this approach to the medium. In the mid 1970s, Myron Krueger developed walk-in, computer mediated, interactive environments as part of his *Videoplace* experience. Within *Videoplace* the position and movement of participants resulted in changes of the images projected on the walls of the space [Krueger].

Mark Bolas’ *Flatlands* example reflects the expanded scope of VR over the 2D plane. In his application, the viewer is positioned in a small cubical room, and can see a representation of Mondrian’s painting “Composition with Line” on two of the walls. However, the two representations differ. One places all the lines of the virtual painting in the planar field. The other representation relocates some of the lines farther away from the viewer than others. The more the viewer moves from the center of the small room, the more they can see the three dimensionality of the second representation.

Example VR applications that fall mostly in the realm of media exploration are *Placeholder*, created by a team by Brenda Laurel (*Placeholder* is discussed in Appendix D of *Understanding Virtual Reality*) as well as Charlotte Davies’ *Osmose*. A primary focus of each application was to

use technology to give participants new (context-appropriate) ways to travel through the virtual spaces. Each application also pushed the method of how those spaces would be seen and heard.

8.1.3 Empathetic experiences

Empathetic experiences are those that use the medium to put the participant “in someone else’s shoes.” Empathetic experiences are commonly presented in novels and motion pictures in which the narrative focuses on a particular protagonist throughout the story. By following the progression of one character, the reader/viewer is guided to strongly identify with the character, and the experiences they are going through.

The narrative of Rita Addison’s *Detour: Brain Deconstruction Ahead* is not as imposing as most novels and motion pictures, but the ability to empathize with the artist’s own personal life experience is very strong. As we share in Rita’s experience, so do we share in the experiences of a great number of others who have undergone similar traumatic events, some of whom we may know well, but had previously not been able to understand how their perception of the world has been altered.

8.1.4 Exploration of the human condition

Subjects such as *exploration of the human condition* include those works that focus on how we as humans relate to one another. These works can help us explore relationships with both our contemporary terrestrial cohabitants, and those that have gone before.

Two examples of such virtual reality experiences are Cirincione et al. *The Imperial Message*, and Benjamin Britton’s *VR Caves of Lascaux* experience. In *The Imperial Message*¹ the participant is put in the situation of progressing from the world of the individual, to the world of the emperor. Along the way they confront many of the conflicts between the state, and the individual [Cirincione]. In *The VR Cave of Lascaux* experience, the participant explores a 3D representation of a real space, but at the same time is made aware that the caves are a place shared by an ancient community, a recent community, the modern world, and even those whom have come to share the VR experience together.

8.1.5 Other and future usage of VR in art

Not all artworks regarding virtual reality are implemented in virtual reality. Some artists choose to express their ideas about the possibilities of VR through other media. This literary study and cultural criticism regarding the positive and negative aspects of VR might be in the form of novels (*Snow Crash*, *The Hacker and the Ants*), rock-operas (*Psychoderelict* and *Lifehouse*), motion pictures (*Lawnmower Man*, *Disclosure*, and *Minority Report*), or any number of other media. All are valid forms of expression, but due to the difference between the medium for discussion and the medium under discussion, it can be hard to accurately convey the issues. VR is an “experiential medium” and requires intimate familiarity to fully comprehend.

¹Not expounded on in this book.

Other artists breach this gap by presenting virtual reality as part of a performance. VR in performance art rides the edge between personal immersion, and detached media study. The audience doesn't experience the physical immersion; the performer does, and the audiences watch their immersive experience. Here the audience gets a better idea of how the virtual environment works, but without the need to don the equipment or learn how to control the environment themselves. The restriction of keeping the audience outside the proscenium prevents them from experiencing the world first hand.

Typically, performance VR uses a large-screen projection to show the virtual world to the audience, along with a view of the immersed participant. The immersed participant themselves may be viewing the world from the same projected image, or may have their own view, such as via an HMD. Two examples of how VR has been used in a performance include Jaron Lanier's musical world VR performance performed at SIGGRAPH 92, and "Machine Child" produced/performed by Robin Bargar and Insook Choi at the Cyberfest² Gala event.

Lanier's world consisted of an abstract environment in which participant/performer travel and world events were controlled via a midi musical instrument interface. Thus as the performer (Lanier) played the virtual wind instrument, the world responded. Held during SIGGRAPH's annual Electronic Theater, Lanier sat before the audience, wearing a head-mounted display, and the audience could see his view on the large theatre screen. "Machine Child" was based on a projection-VR system (the CAVE library), allowing both the audience and participant/performer (Choi) to see the same image. The aural/visual world still only responded to Choi's movements and input controls. "Machine Child" is also interesting in that in keeping with the Cyberfest theme it was a VR experience to explore the world of another medium(s): the world of *2001: A Space Odyssey*. Input focused on different modal controls for the different segments of the narrative: hand gesture, footsteps, and handheld prop.

The advent of the digital computer has enabled artists to explore interactive experiences as an art form. Many of the same artists who showed a previous interest in interactive art who began exploring early digital (programmable) media have now embraced this new medium enabling them to immerse themselves and their audience within another experience. This gives rise to the notion of *experiential computing*. This phrase is particularly meaningful to virtual reality practitioners developing experiences for artistic goals, in addition to trainers and therapists who rely on a realistic experience to have a valid application.

As a developing medium, virtual reality holds great promise for artists looking to pursue their ideas in a medium that provides a means of narrative flexibility and yet can relate an experience into the mind of the participant. As the cost of VR displays has gone down and the ease of creating 3D virtual worlds has gone up, virtual reality has become a much more widely available medium. New venues in public spaces are providing the opportunity to put works of VR art on display.

²Birthday party for HAL at the University of Illinois Urbana Champaign campus.

The Ars Electronica museum for electronic art in Linz, Austria, was one of the first institutions to establish a *CAVE*-projection VR display as a permanent fixture for VR presentations. The *CAVE* is one among many digitally based installations in this museum. By providing such a VR display in a public venue, Ars Electronica provides a conduit for artists working in VR to display their work to a wider public. Each year the museum has hosted a festival including a symposium for the discussion of digital art: the “Ars Electronica Festival of Art, Technology, and Society.”

The future of VR art will also see further development in “new classes” of art. For instance, storytelling, which has mostly taken the form of a one-to-many relationship between storyteller and audience, has now become more of a one-to-one relationship, with greater influence of the audience (i.e., participant) on the events of the narrative. “Interactive storytelling” has been a part of the storytelling tradition since the beginning, but with digital media, it has been expanded into the realm of “personalized storytelling.” The first digital example of this is the famous *Adventure* by Crowther and Woods, in which the participant makes their way through Colossal Cave, with the story seemingly created as they interact with the world (via a text interface).

Now referred to as *interactive fiction* (see Chapter 9), these “text adventures” were very popular during the early period of personal computers. The advent of faster graphics and CD-ROMs has paved the way for fancier interactive storytelling environments, but not always richer stories. The cost of producing graphics for such stories can be very expensive, often imposing limits on the story that the text-based medium does not suffer. Full real-time rendering of 3D scenes and characters will hopefully yield a return to richer, more engaging stories, now as an extension on the tradition of interactive fiction. In fact, the “*NICE*” VR experience (see Appendix A in *Understanding Virtual Reality*) attempted to integrate aspects of interactive fiction into the application, and it is likely that more attempts will be made to include much that has been learned about interactive storytelling by the authors of text-based interactive fiction.

CASE STUDY 8.1: OSMOSE

The *Osmose* virtual reality experience is the result of Char Davies and her team setting out to explore the use of VR as an artistically expressive medium. Davies, a painter who joined Softimage in 1987 as a founding director and vice president of visual research, has a long history of using computer graphics as a means of artistic expression. Consequently, her interests have included developing new mechanisms for expression beyond those provided by conventional three-dimensional computer graphics tools.

A common thread ties Davies’ interest in painting, computer graphics, and virtual reality. She desires to portray

“space as a luminous, enveloping medium” [Davies and Harrison]. The *Osmose* project was a logical step in her progression from static media to interactive and immersive experiences.

From the outset, *Osmose* was meant to provide participants (or “immersants,” as Davies calls them) an experience of solitude. Unlike experiences that pit one participant against another, Davies strove to provide an environment that would allow participants to commune with themselves, to explore their inner self, questioning their everyday assumptions about being in the world.

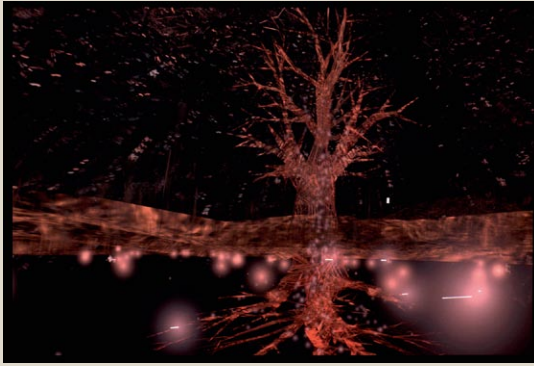


FIGURE 8-2

The worlds in Osmose have a soft feel. This image depicts the clearing world.

Image courtesy Char Davies and Softimage

Application Description

Unlike many other VR experiences, the goal of *Osmose* is not to accurately document or photorealistically portray particular aspects of the natural world. Rather, it is to convey a sense of the ethereal or a sense of being sentient and embodied in a living flowing world. The overall sensibility of the *Osmose* experience was designed to support this goal. The visual aesthetic of *Osmose* is much softer than the gaudy, chunky, cartoon-like imagery of computer games. The style for *Osmose* was derived from Davies's prior work as a painter. Renderings of Davies early work (1990–1993) with 3D computer graphics to create this effect often took as much as 40 hours for a single still image. This frame rate is clearly unacceptable for an interactive real-time application. Rather than achieving acceptable frame rates by reducing image complexity, Davies and her team looked for creative solutions to achieve acceptable rendering speeds, and at the same time presenting imagery that supported the effects and sensibility they were trying to evoke.

The rendering techniques implemented by the *Osmose* team produced objects with varying degrees of realism. Most objects, even the most recognizable ones, still had an element of abstractness associated with them. To achieve this visual ambiguity, some objects were texture mapped with semi-transparent textures. Such use of transparency and “deliberate misuse of the Z-buffer” by the developers

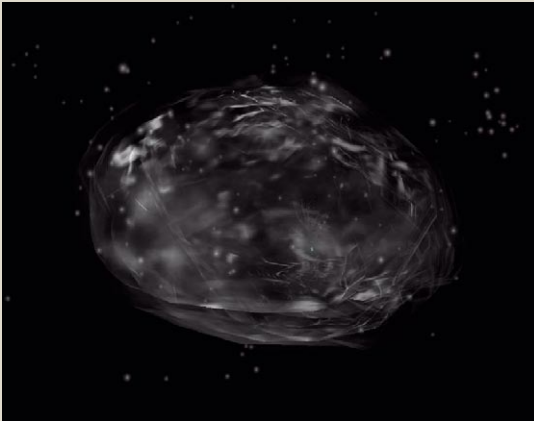
resulted in an ambiguity of spatiality and meaning not often found in computer rendered imagery [Davies and Harrison 1996]. Colors were also carefully modified to support the artist's goal of ambiguity, as well as the desired emotional sensibility of the work.

Some of the visual elements in the application were achieved with particle systems. Paths for particles were determined a priori to achieve the appearance of flows such as streams and insect swarms. For the participant, the culmination of employing all these techniques is the sensation of being embodied yet disembodied while totally enveloped by virtual three-dimensional space.

Rather than using more traditional VR travel interface techniques (such as point-to-fly), *Osmose* required a mechanism for travel that supported the underlying artistic gestalt of the work. Davies' approach to interface design has been informed by scuba diving in oceanic space. Scuba divers control their buoyancy partially through their breathing. The amount of air in the body determines whether they will sink, float, or achieve neutral buoyancy and enable them to hover, perhaps drifting with the current. *Osmose* borrows these ideas, and uses a scubalike travel interface to allow the participant to move among the virtual worlds of the work. In *Osmose*, worlds are located above and below each other, much like the worlds at varying depths in the ocean. In *Osmose*, however, there are subtle boundaries between the worlds, and as such participants may find themselves ascending the sky in one world, and emerging from the ground into the next, or hovering in very ambiguous overlapping boundary zones.

Unlike the more common goal-oriented virtual reality experiences such as mission games, *Osmose* provides a place for inner reflection and questioning. Rather than racing against the clock to beat one's opponent or to see how many objects they can collect or demolish, *Osmose* is meant as a calming transcendental space in which the participant paradoxically feels disembodied and embodied at the same time. Not wishing to objectify the body, there is no avatar representation.

One thing that is noticed when suiting up for the *Osmose* experience is the lack of some of the traditional VR paraphernalia, such as gloves, wands, and other handheld interface devices. The designers of *Osmose* opted to discard

**FIGURE 8-3**

These images show the “subterranean forest” and the “life world.” The grid is the first world that participants encounter and exists solely to allow them to learn to travel through the space by controlling their breathing.

Images courtesy of Char Davies and Softimage

these interface options because they felt that the world was something for the user to control. The interface vest, which tracks breath and balance, is all the control that is provided.

While vertical movement is controlled through breath, horizontal movement control is performed by tracking the orientation of the user’s spine. This allows the user to indicate their desired direction by leaning. The participant’s virtual “buoyancy” was measured by tracking the expansion and contraction of their chest. In addition to reducing the participant’s feeling of “driving” the world, the technology required to implement the experience is reduced by not tracking the participant’s hands or any handheld devices.

The *Osmose* experience also parallels a scuba experience in the manner in which the participant prepares to enter the foreign world. Preparations to enter the undersea world involve donning a mask and buoyancy compensator vest. To enter the *Osmose* world, a participant wears a mask (an HMD) and a vest (to measure their virtual buoyancy by tracking chest expansion and contraction). Hence, by wearing appropriate technology in either case, the participant is able to enter worlds that are otherwise inaccessible.

Osmose consists of a dozen worlds based on archetypal elements of nature [Davies 1995]. Many of the worlds are reminiscent of natural elements such as trees, leaves,

streams, and so on. The first world users encounter is an area in which they can practice the techniques required for traveling in *Osmose*. The practice world consists solely of a three-dimensional Cartesian grid. After practicing navigation, the user encounters worlds that include a central clearing with a tree, a forest, the interior of a leaf, subterranean earth, a pond, an oceanic abyss, clouds, and others. There is also a substratum “code world” that displays the computer code written for the *Osmose* experience. Another area, the superstratum, consists of a “text world” that displays various quotes and poetry related to technology, the body, and nature. Finally, there is a “life world” that is used to bring closure to the experience. Symbolically representing all the other *Osmose* worlds, and all of nature—indeed, life itself—the “life world” appears as a luminous sphere of flecks that fades in all around the user, until finally the world recedes and participants find themselves alone in an empty black space with only a few drifting specs of light, thus indicating that the experience has ended.

Sound is an important component of the *Osmose* experience. In order to attain an organic human nature to the presentation, base samples were taken from a male and a female voice. The samples were then processed to achieve the desired effects. Because they were built on a base of actual voices, the sounds maintained a subliminal connection with the human body presence, a key theme of the work. Some of the sounds in *Osmose* were associated with objects in the environment. Other sounds provided a more ambient quality. Some sounds were dependent on the behavior of the participant. Spatialization techniques were used to aid in the illusion that particular sounds were emanating from the appropriate location. Each world had its own sounds and a haunting melody was introduced in some. Such use of sound not only contributed greatly to the sense of immersion but also amplified the emotional tone of the work. Besides adding to the sense of immersion, the melodies also acted as a wayfinding aid by helping the participant be aware of their general location at any given time.

Thousands of participants have been individually immersed in *Osmose* since its debut. *Osmose* has been displayed in a variety of public venues such as art museums in Canada, the United States, Great Britain, and Mexico. Each participant is allowed 15 minutes in the experience, which was found to be

sufficient to allow the participants to learn to navigate, gain familiarity with the worlds, and “let themselves go” until the appearance of the “life world” signals their time is up.

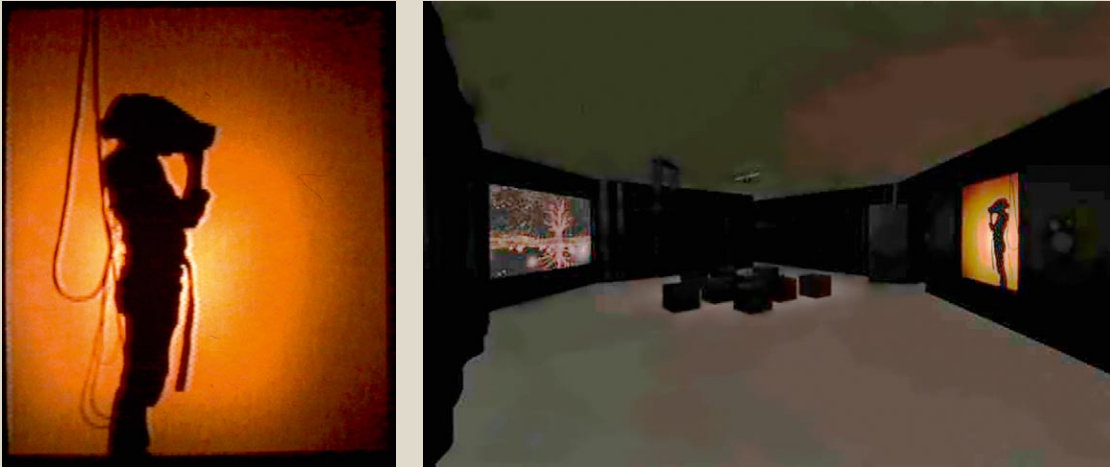
Approximately three people per hour were able to be immersed in the *Osmose* experience. This included the 15-minute experience as well as about 5 minutes to suit up and to remove the VR devices. With only one system available at each exhibition, reservations were taken to help organize the flow of participants. In the event that a person did not show up for their appointed time, an onlooker is selected to take their place.

Because the focus of *Osmose* is to allow people to look within themselves, devices such as projection displays were less appropriate than occlusive HMDs. The HMD effectively shut out the real world, and offered a 100% field-of-regard aiding in the impression that the participant was fully enveloped. Much like visiting the world of the sea necessitates the use of scuba gear, donning a head-mounted display was deemed to be a small inconvenience to bear to enable one to access a new and interesting space. The downside of the HMD was that only one participant at a time was fully able to experience immersion in the work.

To enable more people to get some sense of the experience, Davies added a large stereoscopic projected screen display that could be viewed by audience members wearing low-cost polarized glasses to enable them to see the three dimensionality of the world. To get a good sense of the experience, the large audience needed to see the relationship between the participant’s body movement and the images shown on the big screen. The need for the privacy of the participant required that they were in a separate room where they were alone with an attendant. To balance these two needs, a silhouette view of the participant in the experience chamber was displayed to the full audience. The silhouette was projected on a translucent screen in the installation space along with the rendered world.

VR System

The computing hardware for the *Osmose* experience included a Silicon Graphics Onyx system. For full installations, the Onyx system used three graphics pipes to provide the imagery to the participant and the larger audience. One of the pipes provided the audience view, while each of

**FIGURE 8-4**

To experience Osmose, the participant dons a head-mounted display. The view the participant sees is also displayed on a large screen for others to view. In order to allow the audience to understand the relationship between the immersed participant and the world they are viewing, yet maintain the privacy of the participant, the audience sees a silhouette of the person wearing the head-mounted display and the tracked vest.

Images courtesy of Char Davies and Softimage Inc.

the other pipes provided a single eye view in the HMD. This division of resources generally provided the immersed participant a better frame rate than the theatre audience. The frame rate for the participant was about 20 to 30 fps, while the large audience only saw 10 to 15 frames per second.

A variety of software was used to create and run the *Osmose* application, including *Softimage3D*, and Softimage's *SAAPHIRE* and *DKit* development libraries. Additionally, the team utilized the SGI *Performer* and *GL* graphics libraries. Once the initial framework was established, events within *Osmose* were re-evaluated and rewritten in situ. The creative process thus was the result of "planning and serendipity" similar to how, in painting, previous strokes of pigment can influence what comes next [Davies].

The head-mounted display was a Division dVisor. They selected this model because it had a wide field of view (about 105 horizontal degrees). The developers felt FOV was more important in this application than resolution (345 × 259 pixels). The participant's spine (in two places) and head were tracked with the Polhemus FasTrak™ elec-

tromagnetic tracking system. The scubalike tracking vest was custom built, and measured the expansion and contraction of the user's chest cavity.

A variety of commercial sound products provided the sonic environment for *Osmose*. The Onyx communicated over a serial connection with a Macintosh computer to control two Kurzweil K2000 samplers via the MIDI protocol. Additionally, the experience utilized an Ensoniq DP/4+ processor to provide reverberation effects. The sounds were spatialized using a Crystal Rivers Engineering Acoustetron™.

Conclusion

People who experienced *Osmose* tended to have a very strong response to it. Many wanted to continue longer and felt a sense of loss when the experience came to a close. Some participants, typically very young males, criticized *Osmose* for being paced too slow, with too little to do. Some complained that there was nothing to kill. This is likely a phenomenon due to their expectation that VR games are meant to be high-speed, high-adrenaline, violent games.

Osmose gives a look at a different, very creative use of VR as an artistic medium.

Follow-up interviews with participants who had the opportunity to try *Osmose* both as the immersed participant in the HMD, and as an audience member viewing the large-screen display revealed that the majority preferred the more immersive HMD display. However, the sense of presence of the large screen was quite strong, and was preferred by some participants. In fact, the sense of presence from the projected image was strong enough that the shift in perspective caused by sudden movements of the “immersant” could cause people watching the image to stumble as they stood. The application authors noticed this was more common at the New York installation, where they used a large projection system in a small space.

Osmose was produced by Softimage during 1994–1995. The initial groundwork and testing took a full year to accomplish. World building and other aspects of the immersive component took over 6 months. Since the completion of *Osmose*,

Davies and her team have also constructed a follow-on VR work, called *Éphémère*, which premiered at the National Gallery of Canada in June 1998.

The aesthetic, direction and vision of *Osmose* were provided by Char Davies. Georges Mauro made the models, textures, and animations using *Softimage 3D*. John Harrison provided the custom virtual reality software for *Osmose*, and the sounds were designed, composed and situated in the virtual space by Dorota Blaszczyk and Rick Bidlack.

For its beauty, gentleness, and ability to help us see the world in a new way, CyberEdge Journal awarded *Osmose* with the 1995 Art and Event CJ award.

Through *Osmose*, Davies has produced an experience that intends to “break down the barrier between mind and body leading to an osmotic interplay between sensations of embodiment (due to use of breath and balance) and dis-embodiment (due to being able to float).”

In discussing her quest for a medium suitable to her vision, Davies reveals in her paper “Osmose: Notes On ‘BEING’

In Immersive Virtual Space”: “Osmose is evidence that, after more than a decade of searching, I have finally found the medium capable of expressing my artistic vision” [Davies].

CASE STUDY 8.2: DETOUR: BRAIN DECONSTRUCTION AHEAD

Executive Summary

The idea of *Detour: Brain Deconstruction Ahead* is to allow a person to experience some aspect of the world through the eyes of another. In this example, the creator had experienced a car accident in which she experienced brain damage. Although she appeared completely recovered externally, the accident had affected her vision. It was difficult for her to explain to her friends and family what it was like to see through her eyes, so she worked in tandem with a computer programmer to create a virtual reality application that permitted her to tell her story and to enable participants to see the world as she experiences it.

The application was developed as a *CAVE* application and makes an especially compelling demonstration of the use of head tracking, because the participant is unable to “look away” from the visual anomalies. The use of virtual reality provided catharsis for the creator who was a professional artistic photographer.

Introduction

Detour: Brain Deconstruction Ahead (Detour: BDA) is a VR experience that attempts to communicate the impact of perceptual anomalies that result from brain trauma. In particular, the application simulates anomalies that affect the optical system. Though classified as an artistic experience, *Detour: BDA* is also useful in the medical field to help demonstrate what a victim of brain trauma might be experiencing.

**FIGURE 8-5**

Detour: Brain Deconstruction Ahead begins with the participant in a virtual art gallery. Here, the participant experiences some of application developer Rita Addison's nature photographs. Image courtesy of William Sherman

The creator of the experience, Rita Addison, still recovering from and learning to live with her own brain injury, had a twofold goal for her creation. For her the goal was a means of coping with her own impaired perception and at the same time giving others the ability to “see through her eyes.” Addison’s hope is that a participant will undergo an “undivided moment” in which they become aware of how differently some people perceive the world. By allowing her to give others a taste of what she suffers, and sensing that they are able to comprehend a little of what her life is like, she is more able to come to terms with her own situation.

After her automobile accident in 1992 in which she suffered a brain injury, Addison found that it was difficult to explain to her family and friends exactly what she was experiencing. The only people who understood her situation were others who had also suffered brain injuries of one type or another (accident or stroke victims). Even medical personnel had only a cursory feeling for the difficulties endured by those with these types of afflictions. Addison, and those who had suffered similar trauma as her, felt as though they had “lost their voice” to the rest of the world, since they were only able to find empathy and understanding with other head injury and stroke survivors. This experience was Addison’s way of finding her “voice” again.

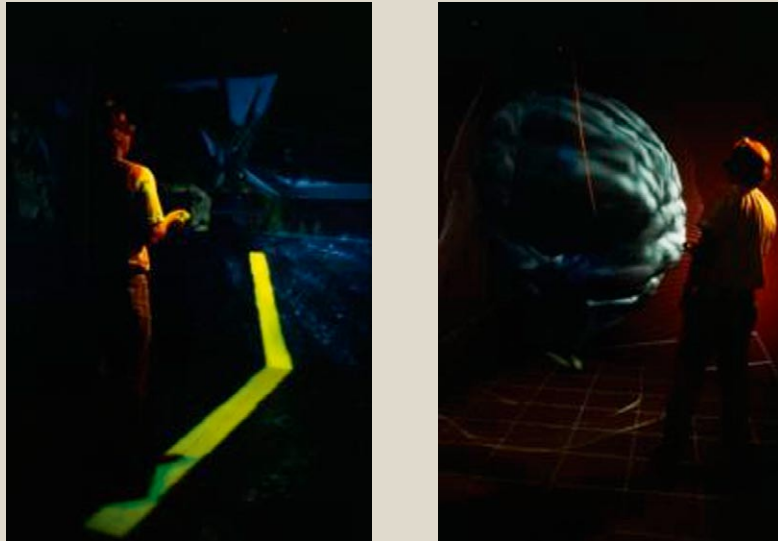
Application Description

Prior to her accident, Addison was an independent photographer, specializing in nature photographs. After her accident, she was no longer able to look through the camera lens, and therefore had lost her ability to make a living in that profession. Thus, she created *Detour: Brain Deconstruction Ahead*. In the first phase of *Detour: BDA*, users find themselves in a Virtual Art Gallery, in which some of Addison’s photographs are suspended along the sides of an infinitely long corridor. The corridor itself is about 10 feet wide, with a marble floor and columns along the edge separating each image. Users are able to travel down the corridor at their own pace, stopping to look at as many photographs as they like. As they continue to move down the corridor, they will find that there are 20 images, which reappear in a never-ending sequence.

The second phase of the experience is a short (about 70 seconds) scripted path of events that represent her accident. The user begins traveling down a divided highway at night, with headlights and signs flashing past until they experience a crash. They then find themselves entering Addison’s brain, watching as the neurons break apart as the voice of a doctor proclaims that “she’s going to have problems.

The third and final phase of the experience returns the user to a corridor of images referred to as “The Gallery of Anomalies.” It is the same corridor, with the same images, but now it runs perpendicular to the original corridor and includes representations of Addison’s personal experience. The participants now find the information presented to their own senses to be filtered by a variety of visual anomalies.

Two perceptual irregularities overlap on the visual feedback to the participant: a limitation of sight mapped to the gaze of the viewer, and a disabling mutation in shape or color mapped to each photograph. Sight is constrained by a filter over the eyes that follows the head motion like a translucent mask that changes its form. This filter changes over time, transforming from hemianopia (tunnel vision), to central scotomata (peripheral-only vision), to clouded vision, and passing through normal (clear) vision back to tunnel vision. The second anomaly is presented through the images on the wall

**FIGURE 8-6**

Left: After a press of a button on the CAVE wand, an accident sequence is triggered in which the participant experiences a late-night crash in an automobile. The consequences of the accident will be experienced later in the application.

Right: As the physicians evaluate the accident victim's condition, the participant is able to walk inside of Addison's brain and witness the snapping and exploding of the nerves that took place in the accident.

Images courtesy of William Sherman

**FIGURE 8-7**

After the accident, the participant is taken to the "Gallery of Anomalies" in which they experience some of the results of traumatic brain injury. In this image, the picture in the gallery of anomalies warps in shape as the participant moves about it.

Image courtesy of William Sherman

of the corridor. Each image has a different anomaly associated with it. That anomaly is activated when the user walks toward and away from the suspended photograph. Some of the images warp space as the user moves in and out, others warp the image itself, changing color or bending lines. Ten different effects are presented this way. For example, an image might bow out toward the user from the center, indicating how one's equilibrium might be disrupted via damage to their vestibular system.

Representation of the Virtual World

The virtual world of *Detour: BDA* consists mostly of very simple objects. This simplicity does not belie the power of the experience. Rather than being constructed in a realistic manner, the world uses abstract imagery to convey the powerful message of what it feels like to live in a world with damage to one's perceptual system.

The marble floor and columns look realistic enough. However, the anomalies are abstractly represented through the warping of space, depth, image acuity, and color of the photographs suspended in the air down the infinite corridor. The automobile accident is also fairly abstract; represented by a dashed median line flashing across the floor as shapes (road signs) pass by. After a short while, one sees (and hears) a crash sending shards of glass off in every direction, reflecting Addison's loss of vision. The user is taken into a Cyberware™ model of Addison's head, going into a brain-shaped object with a simple representation of neurons (as axons and dendrites) exploding and moving toward the viewer as 3D lightning streaks.

Aurally, the user is presented with different sound cues based on what stage of the experience they are encountering. On two occasions, the voice of Rita Addison sets the mood for the galleries. At the beginning of the application she proclaims, "A sense of wonder and enchantment flow through my camera's eye." Later, as the user transitions into the "Gallery of the Anomalies," she can be heard exclaiming, "What's wrong with me? I can't see clearly." During the rest of the time in the galleries, an ambient nature sound plays.

In the accident scene the participant hears the rumble of the automobile engine, leading up to the screeching of the tires, and then the crash. The ambient sound changes as

they enter the cerebral chamber, and the doctor says, "Yes, I think she's going to have problems."

No other senses (haptic, vestibular, olfactory, etc.) are displayed in this experience. Though Addison may have suffered effects to her perception of these senses too, the required extra hardware to display these effects would be too expensive to be cost-effective. In some cases, vestibular effects can be felt based on the warping of space seen in some parts of the "Gallery of the Anomalies."

Interaction with the Virtual World

Detour: Brain Deconstruction Ahead is primarily in the genre of artistic *empathetic experience* that allows participants to view the world from the particular point of view the artist has (or has had) of the world. Interaction is done via the form of a highly constrained walkthrough interface.

More than many other VR experiences, *Detour: BDA* presents itself as a story, with scenes depicting different stages of the narrative. Each scene is demarcated by the visuals and audio displayed to the participant, and signifies a different stage in the story of Rita Addison's own experience as an artist whose ability to work in her medium of choice (photography) ended abruptly with an automobile accident.

Though not explicitly stated, the VR application itself is the next chapter in the narrative. We observe, first hand, where Addison's experience has brought her. It led her to the exploration of a new medium (VR), and her use of virtual reality to communicate her perceptual anomalies.

As mentioned earlier, this experience is a highly constrained walkthrough. In fact, most of the navigation comes in the form of physical movement by the user. During the gallery scenes, the user can move only in the direction of the corridor, pointing (with a wand device) to control the speed. When pointing directly toward the wall of a corridor, the world remains stationary.

The accident sequence is begun with a button press. This is usually done by someone who is running the exhibition, allowing them to gauge when the participant (or group) has exhausted their interest in the normal gallery. After the accident sequence, the participant is automatically placed in the "Gallery of the Anomalies."

The primary objects that the participant can interact with are the photographs in the “Gallery of the Anomalies.” The photographs, suspended in the gallery, deviate in form and color as the user approaches and retreats from them. The shifting in the pictures and geometry represent various neural anomalies experienced by Addison. Anomalies represent such disorders as retinal detachment, hemianopia, scotoma, subdural hematoma, and vertebral artery insufficiency that are brought about by brain injury or stroke.

Computer-mediated interaction with other users or agents is not a part of this experience. The voices of the artist and doctor might be considered a form of agent interaction because they do play a role in the narration of the experience, but there is not any interaction with them.

Detour: BDA is typically displayed on projection VR systems, which allow many people to simultaneously share

DESIGN CHOICE

Rather than subject the participants to complicated interface techniques, the designers focused on communicating an emotional message.

the view of the experience. Interaction among all the viewers present is generally very high. Most participants in this experience are impacted by the experience, and generally immediately begin sharing their feelings with the others in the group.

The world physics of *Detour: BDA* are fairly minimal. Objects are all fixed in place. No collision detection or other interactions between the user and the objects take place, other than the photograph distortions that take place as the user walks about, as described above.

It could be argued that a goal of this experience is a type of *perceptually correct* representation of the world as experienced by brain trauma victims. Though perhaps it could be better explained as being *emotionally correct*, in that it attempts to demonstrate how such sufferers feel as they manage to make their way through the day.

Venue

The original venue the experience *Detour: Brain Deconstruction Ahead* was presented at was the VROOM exhibit at the SIGGRAPH '94 computer graphics conference in Orlando, Florida. This venue, which was organized by the Electronic Visualization Lab at the University of Illinois at Chicago, presented conference attendees the opportunity to see a wide variety of virtual reality applications, ranging from science to art. VROOM had a throughput for each application of five participants sharing a *CAVE* for 5 minutes. Others were allowed to watch in the background. Ideally, a group of up to eight people can fully experience Addison's work in about 10 to 15 minutes.

Since its introduction, *Detour: BDA* has played in many other venues, including Imagina '95 in Monte Carlo, as well as research institutions investigating the use of virtual reality, such as NCSA and EVL.

Addison has also made presentations of and about *Detour: BDA* in other media. Specifically, she presented it in the form of an acceptance speech for winning “Best VR Art experience award for 1994” at the VRAIS '95 CJ awards (CyberEdge Journal) ceremony. Also, Marcus Thiebaut, the programmer of the experience, has created a videotape presentation of *Detour: BDA*, enabling more people to have at least a sample of what the VR experience is like.

The application itself has no mechanisms for enforcing the throughput of the participants. The initial gallery does not take long to explore, and the transition segment runs through a preordained sequence of events, but the length of time to explore the “Gallery of the Anomalies” is indeterminate. Users may wish to explore each of the anomalies in depth, or may rush through the different effects. There is also a significant difference between experiencing the “Gallery of the Anomalies” with full head tracking versus a secondary participant, so *CAVE* visitors will often choose to take turns giving each person an opportunity to empathize with the artist.

VR System

The VR system that *Detour: BDA* runs in is the *CAVE*, a projection visual display environment using Silicon Graphics computers. No input devices other than the standard *CAVE* wand were used to interface with the experience. On occa-

sion, for some venues, a smaller Silicon Graphics computer was used to present a one-screen version of the experience.

Due to her condition, Rita had a difficult time using displays such as an HMD. Therefore, choosing a projection-based display such as the *CAVE* was really the only logical choice. Another choice would limit the ability of the artist herself of being able to experience her own application, which would severely limit not only its cathartic effect on her, but also her capacity of evaluating the work, and sharing it with other participants. While *Detour* has also been exhibited on single screen displays, the surround screens of the *CAVE* are an important element in fully experiencing some aspects of the experience, particularly the tunnel and peripheral-only vision.

Overall, in the opinion of the principal author of the experience, the *CAVE* is the optimal display for *Detour*. Rita considers it to provide “a sense of immersion unequaled by any single-screen display or head-mounted device.”

The programming for *Detour: BDA* was done using the *IrisGL* graphics library on Silicon Graphics computers, combined with the *EVL CAVE* library. The NCSA sound server and library was also used to provide the playback of the voices and ambient sounds heard throughout the experience.

Application Implementation

Development of *Detour: BDA* took place over the 4-month period prior to its debut at the VROOM venue of SIGGRAPH '94. Programming was done almost entirely by Marcus Thiebaut, art student and programmer at the Electronic Visualization Lab at the University of Illinois-Chicago. Marcus was also working concurrently on the *Virtual Director* (Chapter 9) VR application.

The application was developed as a one-time artwork that, once complete, could continue to be experienced by many, but would not continue to be developed further. Development was led by artist Rita Addison, who worked closely with Marcus to shape the final outcome.

Funding and support was provided by Colonel Richard Satava at DARPA, who envisions many useful applications of virtual reality to the medical field, including, in this case, the ability for the experiences of brain trauma victims to be communicated to family (and doctors). Additional sup-

port was also given by the staff and students of EVL, in the way of equipment usage and some programming aid.

Given only 4 months to develop the experience, the choice of designing a simplified world was almost dictated. All of the segments are presented in a stylized manner: the infinite galleries of photographs and the automobile journey down a simple road culminating in the destruction of a work of Rita's art.

The developers chose to use a gallery of Rita's work in her former art form as the focal point about which the experience turns. At the outset, with normal perception the user moves forward down the hall as the undisturbed pictures come into view and pass. Later, everything is turned 90 degrees, forcing the participant to move sideways down the hall, perhaps representing the difficulty by which a brain injury sufferer goes through life.

Conclusion

Though no experimental analysis was done, there are many anecdotal accounts regarding the experience. Of particular interest are those from the family and friends of people who have suffered similar effects to those portrayed. The consensus is that they have been given a rare glimpse of insight inside their loved ones' trauma, and a better understanding of the person.

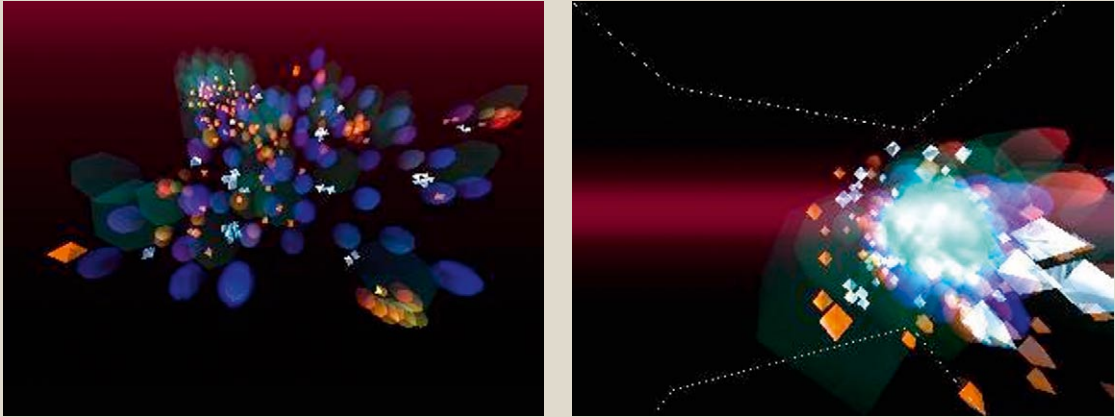
One of the big accomplishments of this application was its demonstration of how VR can be used to enable one person to experience the world of another. Even for those who do not personally know someone who suffers from brain trauma, *Detour: Brain Deconstruction Ahead* is often singled out as a moving VR application.

It was also singled out by the Cyberedge Journal in 1995, when it announced their annual award for best application of VR of 1994 in the Arts and Events category:

For creating a work of beauty, intensity, and long-lasting value, we are proud to have presented Rita Addison with the first CJ for Arts and Events!
[CyberEdge Journal—Virtual Reality Products]

The goal of allowing others to experience Rita's perception of the world has definitely been attained.

Because virtual reality deals with modifying the environmental information provided to the senses, it is an excellent

**FIGURE 8-8**

Addison's VR application *Synesthesia* represented sense information as different forms.

medium for experiences such as Addison's to demonstrate an altered perception of the world. Many of the effects suffered can only be properly experienced in a medium that tracks the movement of the viewers head and body.

The anomalies associated with the gaze direction of the tracked participant, such as tunnel vision, peripheral vision, and others, are particularly effective for the viewer who's gaze controls where the blind spot will be. For the other viewers, who can move their heads to see around the blind spot, the effect is much less immersive. The anomalies that are affected by user movement in the space are much more

easily understood by the nontracked participants (assuming the "driver" does indeed move around, triggering the effects).

The major discoveries made by Addison were the power of VR to allow participants to free themselves from daily concerns, and in many instances, strongly impact the emotions of the participants. At the first public showing of *Detour: Brain Deconstruction Ahead*, Rita felt the emotions rise in the people around her. She knew that she had succeeded in what she had set out to accomplish.

Next up for Addison was her project *Synesthesia*, which is more in the VR art genre of *exploring the medium*.

though it continues to have elements relating to how people with a disorder (Synesthesia) experience the real world. Sufferers of this disorder perceive information across senses; for example, music is seen, tastes are experienced as geometrical forms, etc.

Finally, perhaps the biggest accomplishment of the *Detour: Brain Destruction Ahead* VR experience is its role as a cathartic agent for Rita Addison. In her article in the journal *Computer Graphics and Applications*, she says:

Detour became a vehicle of profound catharsis for me. It helped me accept and adapt to chronic impairments and fight my way through depression.

CASE STUDY 8.3: THE VR CAVE OF LASCAUX EXPERIENCE

Executive Summary

The VR Cave of Lascaux is an interesting combination of science and art. The application is an artistic recreation of the cave of Lascaux, but while many aspects of the experience are scientifically accurate, the goal is not scientific realism but rather a way to experience the magic of the cave. The application is typically experienced in a head-mounted display, and navigated with a six-degrees-of-freedom input device. This experience was developed as a labor of love and represents the collective efforts of the artist and many volunteers. A number of magical elements are present in the experience, such as the cave paintings that “come alive” with animation when the participant gazes intently on them.

Introduction

The cave of Lascaux is a treasured monument to a past culture. Just as importantly, however, it represents how humans relate to one another. Led by his own fear of the future, the potential damage the human animal could cause to its world, and the world of other creatures, artist Benjamin Britton looked to another time for clues to the human condition. He looked to the Ice Age, which was a time in which human culture had been decimated. His research brought him to a shrine of an ancient culture, a cave with historical as well as magical significance brought out by the paintings of a past culture. Britton decided to create an electronic version of the cave of Lascaux. The focus would be to highlight the communal aspects of the cave, and bring forth the opportunity for museum goers to participate in a modern day communal event with the cave as the basis. Out of this work came the *LASCAUX* experience.

The cave itself was lost shortly after the last ice age, and rediscovered in 1941. After the Second World War, it was opened to the public. The tremendous traffic of people who came to experience the cave caused it to deteriorate from the resulting increase in moisture and bacteria brought into the cave. To solve this problem, a replica of the cave



FIGURE 8-9

Texture maps are used to create the visual look of the VR Cave of Lascaux. This is a view in the center of the Fresco in the Hall of Bulls.

was constructed nearby. However, even the replica had to be closed due to the ill effects of human traffic. Now, only a select group of people (mostly serious researchers with a compelling need) are allowed access to the cave.

The Ministry of Culture of France considers the cave a rare cultural treasure, and they do not wish to see the cave lose its spirit and message through commercialization. However, they were interested in Britton's work in digitally capturing the essence of the cave, because it respects the cave near Lascaux, and the Ministry has supported his work through the provision of information about the cave. With their help, Britton and his collaborators created a virtual reality experience of the Cave of Lascaux.

This experience was created as a work of art intended for a contemporary art audience. While at first blush it appears to be done as a historic travel application, the artist is quick to point out that scientific and historical accuracy was not a primary goal. Hence, this piece is an artwork, intended to communicate not just about the physical cave but, indeed, the nature of humankind. It is for audiences to

³In fact, before the cave was closed, the local population held occasional parties and other personal gatherings in the cave.

**FIGURE 8-10**

The exterior of the cave was modeled at a lower level of verisimilitude than the interior. Here one can see the entrance to the cave.

**FIGURE 8-11**

A view of the stylized representation of Montignac, France. From here, one can fly in “Peter Pan” mode, and explore the “Cosmos” if they desire.

learn about today’s world, how it relates to the past, and how it might relate to the future.

The physical cave that inspired this piece serves as a record of the populations that inhabited the area through time. Each culture has altered the cave, and the cave has altered each culture, including 20th-century man. Recent marks to the cave include the addition of electricity to some areas of the cave, as well as the deterioration of its content by the masses who wished to experience the cave first hand, and later (in the 1950’s) by the vault-like door added to the exterior to control access.³

The design of the door itself became controversial because it was designed in a Mycenaean style of construction, and critics were concerned that future archaeologists would therefore think that the cave’s door had been built during the Bronze Age.

The cave of Lascaux was chosen as the setting for Britton’s VR experience because of its nature as a special place for those who have experienced it. From the outset, a goal was to maintain a sense of the cave’s mystique. Thus, even the VR experience is available only in certain limited venues. Britton believed that the venue and social interactions of the participants make an immense contribution to how the

work is perceived. People will come because of a general interest in the history of mankind, the meaning of human existence, represented by the cave. Donning the HMD to see the cave is only a part of the participant’s experience. The procedure of dressing up to visit a museum, queuing at the entryway, rubbing shoulders with neighbors, and even the effort of driving to the museum and parking contribute to a person’s overall experience with the application, and the human community.

Now that the real cave is closed to most people, this is also the best way for people to have a taste of what walking through the real cave is like. At the same time, it potentially reduces the travel time and expense from that of a trip to southern France, to that of a trip to a nearby museum, although Britton states that a trip to the Dordogne region around Montignac and the real cave is well worthwhile.

Application Description

The *LASCAUX* experience replicates, in a virtual world, many of the features of an existing location on the earth. The fragility of the site means that only a few people are allowed access to that location. Yet, natural curiosity regarding the peoples of the past created a need for an alternate way to explore this site.

The virtual world consists primarily of an interior model of the actual cave, including walls that are texture mapped with images of the actual cave walls. Outside the cave the participant can explore a small region of the surrounding French landscape (including a stylized view of the nearby village of Montignac) and the “Cosmos.” The “Cosmos” refers to the world beyond the perceivable, the universe. If the viewer chooses to explore not the cave but the valley surrounding the cave, they can fly “like Peter Pan.” If they fly up and into the sky, then they can fly *through* the sky and into a sort of outer space environment from where they can see the valley far below them, and the whole virtual world. This interpretation of the “Cosmos” was inspired by a 14th-century woodcut of a shepherd who climbed to the top of a mountain in order to pierce through the sphere of the heavens and witness the working of the “kosmos.”

Representation of the Virtual World

Because *LASCAUX* is a work of art, a conflict of interpretation versus verisimilitude arises. How much of the work represents what the artist wants to convey, and how much is an attempt at a realistic rendering of the object of study? *LASCAUX* is a representation of an actual entity, yet the reconstruction will have impressions left by its creator. In Britton’s words: “It will have a veil.” His goal was to strive for the highest verisimilitude attainable, making the veil as light and transparent as he could.

With realism as a general representational goal, a survey of the shape of the cave interior along with many high-resolution photographs were used to create the virtual cave. The French government allowed Britton to make use of government data and photos for his project, thus re-enabling people to experience a likeness of the cave.

Not all aspects of the VR experience were designed to be realistic. The re-created cave also has some magical qualities. The participant might trigger an event relating the pictures on the walls. For example, a video representing more recent human history can be triggered by certain circumstances in the cave. Or perhaps, the user might notice subtle changes in the cave; finding new items in a previously visited, formerly empty, area.

Internally, the world is stored as a polygonal description of the cave surfaces plus thousands of detailed texture

maps, covering nearly the entire cave. The video events are stored as full-motion video files that are loaded into the system at appropriate times.

The resultant experience is a visually engaging world, with the same shape, passages, and wall paintings as the real thing. Taking advantage of high-resolution imagery scanned from the walls of the actual cave, combined with accurate information about the physical shape of the cave, the interior of the cave is very visually realistic. The exterior of the cave is more of a stylized representation, with an icon view of Montignac nestled in the hills of the Dordogne region. During some sequences of the experience, captured video replace the center of the viewer’s vision, showing related images from a more recent era.

Although the experience is decidedly visually focused, there are some audio elements as well. Most of the sounds heard by the immersed participant are long ambient soundtracks mixed together partially based on their location in the virtual world. The ambient sounds include a Chinese flute song titled “Autumn,” and several sounds recorded in Kentucky’s Mammoth Cave. A flute sound is heard when the listener is outside the cave, and inside, a mixture of various sounds clips of Mammoth Cave are heard. A few other sounds such as door squeaks and bursts of wind are immediately played to correspond with appropriate events.

Because the ambient sounds were long, immediate response to movement throughout the cave was not practical. Instead, sounds were faded in and out and mixed together as the listener moved through the space. This produced an effect that Britton considers to be a *mise en scene* produced by serendipity as the program selects different sound clips.

For various reasons, including expense, no haptic or other sensory feedback is part of the *LASCAUX* experience.

Interaction with the Virtual World

We have classified the *LASCAUX* experience as part of the genre in which the *human condition can be explored* through virtual reality, although it also contains elements of other genres, such as the historical or travel walkabout.

The basic form of interaction between a participant and the experience is traveling through the space using a typical *desktop 6-DOF input device*. The *walkthrough* form of

**FIGURE 8-12**

Some imagery is associated with magical events. For example, staring at these Bison triggers the play of a video clip in the cave.

travel was chosen because of how closely it resembles the physical experience of visiting the actual cave. By putting pressure on the input device, the virtual world responds by moving the user to a new location in the world.

With exploration of the locale as the primary function of the experience, the narrative was loose. As the user travels through the cave, they can stop and ponder various paintings or other features they may come upon. Occasionally, pondering a particular image on the wall (i.e., staring at it) will cause the image to transform to a moving image. Watching the moving image might conjure up thoughts on the relationship between the world they are currently experiencing with a different time and place. As the video plays, the travel controls are disabled.

As the user travels outside the cave, they see a stylized representation of the nearby village of Montignac, France. From there, they can fly to the “Cosmos” for an over-arching view of the world.

Navigating through the world is quite simple. The participant controls travel using a standard desktop 6-DOF device such as a Spaceball™ or a Magellan™ (aka Spacemouse™). Inside the cave, the user is constrained to move only at a fixed distance above the floor. Pressure on the device in any direction moves the user in that direction. Buttons on the device are also available to cause a jump to a pre-

**FIGURE 8-13**

Although the experience in the cave is one of solitude, certain actions remind one that they are not alone in the world. This image shows a table and chairs set up in the main hall. These objects were not there when the participant first visited the hall. They appear only on subsequent visits.

defined specific location, or to give a predefined guided tour through the cave. However, since the buttons were not visible when wearing the HMD, they would typically be used by a guide staffing the exhibit.

There are no agents per se in the application, although one might get the feeling another entity is a part of the world when they return to the main hall to find a table and chairs set up for a dinner party. Also, the “hot paintings” sense when a viewer is concentrating on them, and spring to life with a video animation have an agent sort of aspect to them.

Events such as the “sudden” appearance of a mysterious table are what Britton refers to as “dormant events.” These events are only activated once the participant has visited certain locations, setting off a tripwire. The tripwire can be triggered by very specific circumstances: the time a viewer spends in a certain space, for example, or even standing in specific locations or looking in certain directions.

Because it is designed to foster a communal experience, one might think it is desirable to enable multiple users to simultaneously be able to share the experience first hand. The artist, however, considers the balance between the value

of shared experience against the personalized sense of being alone and in contact only with the cave. Each viewer wishes to feel themselves connected, and in a sense, sharing the experience can, for some viewers, dilute the experience



FIGURE 8-14

The venue allows people to see what is happening.

Photo by Lisa Britton

and make it less “their own.” From a more practical point of view, it is also uncommon for a museum installation to have more than one virtual reality system. Thus, for both these reasons, the added software complication was not considered worth the effort.

The physics of this world are fairly simple and deal entirely with concerns of how participants travel through the world. Since there are no objects in the world that the user can manipulate, there is no need for gravity or other worries such as collision detection between objects. *LASCAUX* uses a terrain-following method to simulate gravity, and has a “Peter Pan” mode allowing the participant to fly around the valley and off into outer space. There are also secrets in the cave that erase all “virtual physics and leave the participant immersed in virtual space without ground, and in the true zero gravity of basic VR.”

Venue

The *LASCAUX* experience was specifically designed to be shown in a community museum venue. The artist views the most appropriate venue as the art museum, rather than

⁴ *The World-Wide Web.*

⁵ *Internet Service Provider.*



FIGURE 8-15

The museum venue allows onlookers to see the technology being used by the immersed VR participant.

Photo by Lisa Britton

science museums, though under the right circumstances, science museums might be appropriate. The goal was not as a scientific lesson about the cave, but rather a study of human culture, and the nature and history of humankind. The application loses much of its point and importance when one scrutinizes it for scientific and historical accuracy (although even in scientific forums, *LASCAUX* is a prime example of excellent 3D graphics and site reconstruction).

The experience is not widely available, and can be viewed only in specific carefully controlled venues. This helps to maintain the mystique associated with the actual cave of Lascaux. In Britton's words:

I remain concerned about the tendency of the WWW⁴ to present viewers with trivialized, disposable, Kleenex-culture experiences, and wish for viewers that their experience of LASCAUX can be richer than a mere mouse-clicky passthrough while logged onto their ISP⁵. For this reason, I raise the bar and make the experience require a higher level of personal investment from audience members. A viewer must really wish to see the virtual cave if they are to make that experience happen for themselves.

Venues at which *LASCAUX* has appeared include museums in Paris and Montpellier, France, Tokyo, Italy, Austria, and Korea, the Epcot Innoventions venue in Disney World,

SYSTEM CHOICE— DISPLAY PARADIGM

A head-mounted display was selected as the display paradigm, specifically the Virtual I/O unit an inexpensive model (less than \$1000) that includes a system for orientation tracking.

Florida, as well as several appearances at the SIGGRAPH conferences on computer graphics.

The application was designed to allow the participant to go on a self-guided tour of the cave. In the initial exhibitions, however, Britton preferred to have it hosted by a guide. This is helpful for getting each participant situated, as well



FIGURE 8-16

A museum visitor wears a head-mounted display, and navigates through the VR Cave of Lascaux.

Photo by Lisa Britton

as being helpful with the flow of traffic through the exhibit. After watching many people use his application, Britton has learned that participants are able to figure out the system on their own. In a museum setting, the general length of time for being immersed in the experience is about 15 minutes.

VR System

The intended venues for the *LASCAUX* experience provided the primary constraint on the system design: low cost. Art museums frequently spend on the order of

DESIGN CHOICE

The combination of a monoscopic image without translational tracking was selected for this system for better user comfort and lower system cost.

\$50,000 for an exhibit. In the early 1990s it was more challenging to meet this constraint for a virtual reality-based experience. In fact, the cost of producing a VR experience is generally much higher than this, and artists working in

such a technologically based medium can find it difficult to have access to the latest technology. The cost of producing *LASCAUX* was about \$500,000. Individual installations of the exhibit can cost as little as \$25,000.

Britton felt that by choosing an HMD display, people would be more intrigued by the application and, compelled to give it a try. The Virtual I/O system chosen also had the attributes of being light weight, inexpensive, and included its own tracking hardware (further reducing overall system cost).

Onlookers can get a view of what the participant is see-

SYSTEM CHOICE

Computer System

A dual processor Intel-CPU computer running Windows NT and Intergraph graphics processors with 64 megabytes of texture memory was chosen as the computer platform to run the application.

ing on nearby display monitors. In fact, multiple monitors can be arranged to show a much wider field of view to the nonimmersed viewers than what the participant experiences in the HMD.

The tracking system on the HMD uses accelerometer technology. Thus, it only tracks orientation (not translation) and requires frequent recalibrations. The recalibrations can be handled without too much trouble by a guide staffing the exhibit. The lack of translational tracking is less noticeable because monoscopic images were chosen over stereoscopic images.

SYSTEM CHOICE

Input System

A Spaceball™ 6-DOF input device or a standard mouse is used to allow the user to move through the virtual world.

Though choosing to display monoscopically and without respect to head translation lessens the depth cues available to the participant, the designers felt the benefits outweighed this. The reason was not just because of computational and hardware costs, but also because of user comfort. Discomfort in the form of nausea is occasionally a problem in HMDs. This is generally caused by system lag and improper align-

ment in the stereoscopic imagery. By choosing to display a monoscopic image, alignment headaches are eliminated. While there will always be some lag in the system, the use of accelerometer based tracking reduces the amount of display lag in the tracking component.

The compute and graphics system was chosen because it met the financial goal, yet is also quite capable of handling the computational requirements of the application.

Most of the audio was handled through the use of normal audio CD tracks. By using an audio CD, the developers could eliminate the concern of consuming large portions of memory for high quality sound samples. Mixing of different tracks, with a few shorter sound samples stored in RAM was handled by the main application.

Six degree-of-freedom input devices are typically intuitive to use for the average person, and were selected as the primary input devices.

More recently, the experience has also been publicly displayed in a projection-based (“CAVE-like”) display system at the 1997 SIGGRAPH computer graphics conference. In this showing, viewers were not given the opportunity to control the movement through the cave, but the wide field of view provided by the surround screens was very impressive. When run in a CAVE environment using Silicon Graphics computers, stereoscopic imagery is presented.

Application Implementation

The original concept for this artwork began as a video wall display run by several laserdisc players. Later, a switch was made to a virtual reality interface. This switch in media led to multiple benefits. A VR interface allows the work to become much more interactive, letting participants move through the cave with total freedom. Plus, the added intrigue associated with virtual reality, would help to generate interest in the artwork. And finally, a small VR System (PC and low-cost HMD) is much easier to install into new venues than a video wall with many large monitors, each with a laserdisc player, plus a control computer.

Work began on this application on an IBM-compatible 386 PC in 1991. The creation of the full application went on for 5 years. During that time, it was exhibited on a few occasions as a work in progress. In 1996 work was completed, and with the possible exception of improving hardware, it will remain

as it is. By 1998, the typical system used is an Intel Pentium running at 300MHz and including MMX enhancement.

Although the original implementation utilized an HMD, they have also implemented a nonimmersive PC based multi-screened stationary display, providing a wide field of view. They have also demonstrated the experience running on a CAVE-like projection-VR display using Silicon Graphics computers.

The application was developed by a core team consisting of University of Cincinnati Professor of Art, Benjamin Britton, a technical consultant, and three graduate research assistants. Most of the funding came from the University of Cincinnati, the Ohio Arts Council, and the Chicago Center for New Television consultancy. Total cost to create the application was about \$500,000. However, this does not fully reflect the amount of person-hours used to create the application. Much of the work was done for the love of the art and/or opportunity to work on an exciting VR application. Additionally, Intergraph Computer Systems donated machines for production and exhibition in support of this project.

Core tasks of designing and creating applications were handled by different individuals of the design team. One research assistant was primarily a texture map specialist and did the work, unpaid, as an independent study for college credit. Another research assistant focused on the hardware, modeling, and graphics. The remaining student was described as "rationally minded and a logical thinker." He focused on data management and other logistical details.

The technical consultant, David Britton (the artist's brother), focused mostly on the development of an authoring system, used to create the final application. There were many other tasks that were performed by about 15 volunteers during the height of production. They worked (often around the clock) to digitize the cave images, create the video clips, mix the sounds, etc.

Many tools were used to bring together this application. In addition to the custom made authoring system, Sense 8's 3D virtual world package, *World ToolKit™*, was used as the world management and rendering system. Adobe *Photoshop™* was used to montage many of the scanned cave images together. These images were then texture mapped onto polygons. This work involved tweaking the correspondence between the polygons and the textures by hand. Microsoft *Visual C++*

was used to create the custom VR authoring tool and engine for the project.

Conclusion

The artist's goal of *LASCAUX* was to bring people "into personal contact with part of our common cultural legacy as members of the family of humanity." Successfully accomplishing such a goal is very subjective. One method Britton used to determine if he was reaching the audience was to mingle with the patrons gathered at the opening of an installation and watching as they participated in the experience. In France, he found the people particularly enthusiastic about wanting to see the project, with waiting lines "around the block outside the Le Monde de l'Art galerie in Paris."

While not a life-saving type of application on the surface, it has helped to provide an opportunity for many other people to experience the otherwise restricted cave of Lascaux, allowing them to have their own encounter with the mysteries of the cave.

The experience has been successfully exhibited at several venues, providing many more people an experience of the cave of Lascaux. Also, it has generated considerable interest, with TV coverage by CNN and other networks. Britton reports that even the number of people who have had the experience is quite large. At least 12 million viewers in galleries and museums on four continents since its 1995 premiere in Kwang-Ju, Korea, have tried the experience. This includes about 10 million people who ventured into the Innoventions Pavilion at Epcot Center in Disney World, and another 2 million at a 4-month show in Paris. Only a fraction of these had the opportunity to actually experience *LASCAUX* first hand, of course. In addition, hundreds of millions of viewers have seen television coverage on CNN, NHK, BBC, Canal+, and Korean, Italian, Spanish, Portuguese, and Dutch television stations.

Another significant accomplishment of the application is that it is a good example of successful use of technology by an artist. The artist became intimately familiar with the technology involved, and also developed new technology in support of his goals. As such, it was recognized by *Discovery* magazine as a recipient of a 1995 Discover Award in the "Sight" category.

The performance of the application was very good, giving the participant good visual cues of being in the cave environment. Even in the early systems, the only noticeable delays were during the loading of the embedded video clips. After stepping up to 300MHz Intel Pentium CPUs with MMX enhancements, the video as well as the interactive graphics perform at real-time frame rates.

Though, it is perhaps too strong to say that it feels like you are actually exploring the real cave, that is not entirely the goal. One can easily become caught up in the exploration, and magic of the virtual world. On the other hand, when rendered in a CAVE-like visual VR display the feeling of being in a real place is significant.

There are two important points that are highlighted by the work done in the creation of this experience. One is that when working with technical equipment in the creation of art, artists must learn how to balance the amount of control given to the participant versus what they are *not* allowed to do. In most media, artists are accustomed to controlling the entire presentation. However, in media that allow significant user interaction, they must allow the user to make choices, to have (some) control over the overall experience.

Britton believes:

With interactive composition, artists must anticipate viewer response and develop a meta-AI which communicates to the viewer an attitude and a sense of control. Artists do not relinquish control to viewers in interactive art, on the contrary they dangle the illusion of control in front of the viewers, inspiring and tempting them to explore and invest their wills in the interactive art experience. Only a poor interactive composer would give up control, except as a ploy to make their audiences feel particularly empowered at some point. In interactive media, no less than in any other medium, artists control the entire presentation and provide their audience only with the semblance of control. Empowering viewers is a technique for comforting and entertaining them, not a technical necessity.

Another important point applies to work being done on computer-based historical reconstructions. As with the



FIGURE 8-17

The artist, Benjamin Britton, experiences his own work.

LASCAUX experience, where the goal was to minimize the veil between the physical cave and how it is presented in the HMD, anyone working on reconstructions needs to recognize that they *cannot* tell the whole truth. In other words, all media place a veil between what is represented and what is perceived, and it is important for content creators (and viewers) to realize this fact, especially when creating an experience that purports to accurately represent how the real world existed at a previous point in time.

Britton found that the context of a museum gallery was an ideal place to present art to an audience. In Britton's opinion, VR displays of this nature are well suited to display in a museum because it provides a physical space in which one can manifest a virtual environment. He believes that the use of less-encumbering immersive technology, such as Projected-VR displays, provides a more natural and inviting space, but he also believes that the flat screen—like the book, the window, and the painting—is a valuable display genre that will never disappear and will remain a vital part of virtual reality presentations.

Also, one must be aware of the clash between VR hardware that is powerful but expensive and hardware that is less expensive but can do the job. This is one battle Britton has engaged himself during the creation of the *LASCAUX* experience. He deliberately chose to use a PC system in order to make his experience affordable to museums. To